

D: Discrete Catalog

Source file name: `discrete.c`, `discrete.cpp`, `discrete.java`, or `discrete.py`

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In a certain country, natural disasters strike often enough that the authorities must constantly evaluate their preparedness. Earthquakes, floods, and landslides regularly disrupt local communities by isolating sites and cutting off access to food, water, and other vital supplies. To plan for these crises, the national crisis center runs simulations on a map of the region.

The region under study has n sites labeled $1, 2, \dots, n$. A *scenario* describes which sites are disrupted in a disaster. Formally, a (n, k) -scenario is a tuple $[a_1, a_2, \dots, a_k]$ of pairwise distinct integers in the range $1..n$, written in *strictly increasing order*, i.e., $1 \leq a_1 < a_2 < \dots < a_k \leq n$. This ensures that each group of k sites is represented exactly once in the catalog.

To manage these simulations, the crisis center maintains a *Crisis Catalog* that assigns each possible (n, k) -scenario a unique *ID*. All scenarios of size k are listed in *lexicographic order* and numbered starting from 0. In lexicographic order, one scenario comes before another if, at the first position where they differ, the site number in the first scenario is smaller. This is the same way words are ordered in a dictionary, but applied to integer sequences instead of letters.

For example, there are 10 different $(5, 3)$ -scenarios, with IDs in the catalog as shown next:

$(5, 3)$ -scenarios	ID
[1, 2, 3]	0
[1, 2, 4]	1
[1, 2, 5]	2
[1, 3, 4]	3
[1, 3, 5]	4
[1, 4, 5]	5
[2, 3, 4]	6
[2, 3, 5]	7
[2, 4, 5]	8
[3, 4, 5]	9

Given n , k , and a specific (n, k) -scenario, your task is to determine its ID in the Crisis Catalog.

Input

The input consists of two lines. The first line contains two blank-separated integers n and k , with $1 \leq n \leq 60$ and $1 \leq k \leq n$, indicating the number of sites and the size of the scenarios. The next line contains a list of k blank-separated integers $a_1, a_2, \dots, a_k$ satisfying $1 \leq a_1 < a_2 < \dots < a_k \leq n$.

The input must be read from standard input.

Output

For each test case, output a single line with the ID of the (n, k) -scenario $a_1, a_2, \dots, a_k$.

The output must be written to standard output.

Sample Input	Sample Output
5 3	0
1 2 3	4
5 3	7
1 3 5	9
5 3	0
2 3 5	1
5 3	2
3 4 5	484650
3 2	75363982551
1 2	
3 2	
1 3	
3 2	
2 3	
32 6	
4 8 16 17 25 31	
60 10	
31 32 33 34 35 36 37 38 39 40	